

Unnamed_Unit

Unit 5 - Knowledge Check

PDF Documents

PDF Document 1: 1738145858092-Chapter 6_unit 5 - Knowledge Check.pdf

This PDF document is included in the unit materials.

KNOWLEDGE CHECK

1 What is the principle behind the operation of traffic lights using different colors?

- ☐ A Color absorption
- ☐ B Color reflection
- ☐ C Wavelength variation
- ☐ D Intensity modulation

CORRECT ANSWER ☒ C

2 Which color has the shortest wavelength in traffic lights?

- ☐ A Red
- ☐ B Green
- ☐ C Blue
- ☐ D Orange

CORRECT ANSWER ☒ B

3 Why are green lights used to signal 'go' in traffic systems?

- ☐ A Because green light has the longest wavelength and is easiest to see.
- ☐ B Because green light has the shortest wavelength.
- ☐ C Because green light uses the least energy.
- ☐ D Because green is a calming color.

CORRECT ANSWER ☒ A

4 Which programming structure would you use to change light modes based on ambient light conditions detected by sensors?

- ☐ A Nested-If
- ☐ B Switch-case
- ☐ C For loop
- ☐ D zNone of the above

CORRECT ANSWER ☒ A

5 What would happen if you used an 'or' logical operator instead of 'and' when programming a traffic light system to check multiple light conditions?

- ☐ A The system becomes more restrictive.
- ☐ B The lights will change only when all conditions are met.
- ☐ C The lights may change when any single condition is met.
- ☐ D The program will not compile.

CORRECT ANSWER ☒ C

6 How can variables be used in programming the sequence of a traffic light?

- ☐ A To store the current color of the light.
- ☐ B To count the number of cars passing.
- ☐ C To calculate the speed of passing cars.
- ☐ D To measure the width of the road.

CORRECT ANSWER ☒ A

7 WWhat programming method would be best for implementing a countdown timer in a traffic light system?

- ☐ A Variable assignment
- ☐ B While loop
- ☐ C Recursion
- ☐ D Use of an external library

CORRECT ANSWER ☒ B



Critical Thinking Challenge

- 1 Consider how understanding the wavelength and intensity of different colors used in traffic lights can help improve their visibility in various weather conditions. Propose modifications that could enhance traffic light effectiveness during fog or heavy rain.

- 2 Design a traffic light system that uses a combination of light colors and patterns to communicate more complex traffic management instructions, such as variable speed limits or emergency vehicle priorities. Describe the types of light waves and the programming logic you would use.